Decimals - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

Work out $4.7 + 2.35$

$4.7 + 2.35 = 7.05$ (line up the decimal points)

Answer: 7.05

Question 2 *Practice question written for this worksheet (not from a past paper)*

Work out $12.5 - 4.78$

$12.50 - 4.78 = 7.72$

Answer: 7.72

Question 3 *Practice question written for this worksheet (not from a past paper)*

Work out 0.8×0.6

$8 \times 6 = 48$, 2 decimal places $\rightarrow 0.48$

Answer: 0.48

Question 4 *Practice question written for this worksheet (not from a past paper)*

Write $\frac{5}{8}$ as a decimal.

$5 \div 8 = 0.625$

Answer: 0.625

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Write 0.35 as a fraction in its simplest form.

$0.35 = \frac{35}{100} = \frac{7}{20}$

Answer: 7/20

Question 6 *Practice question written for this worksheet (not from a past paper)*

Work out $3.2 \div 0.4$

Multiply both by 10: $32 \div 4 = 8$

Answer: 8

Question 7 *Practice question written for this worksheet (not from a past paper)*

Work out 1.5×2.4

$15 \times 24 = 360$, 2 decimal places $\rightarrow 3.60$

Answer: 3.6

Question 8 *Practice question written for this worksheet (not from a past paper)*

Work out 0.15×0.36 without a calculator.

$15 \times 36 = 540$, 4 decimal places $\rightarrow 0.0540$

Answer: 0.054

Question 9 *Practice question written for this worksheet (not from a past paper)*

Work out $6.4 \div 0.16$

Multiply both by 100: $640 \div 16 = 40$

Answer: 40

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Question 10 *Practice question written for this worksheet (not from a past paper)*

A bottle holds 1.25 litres. Glasses hold 0.05 litres each.

How many glasses can be filled from the bottle?

$1.25 \div 0.05 = 125 \div 5 = 25$

Answer: 25

Question 11 *Practice question written for this worksheet (not from a past paper)*

Write these in order, smallest first:

$0.45, \quad \frac{3}{8}, \quad 0.405, \quad \frac{7}{16}$

As decimals: 0.45, 0.375, 0.405, 0.4375

Order: $\frac{3}{8}$, 0.405, $\frac{7}{16}$, 0.45

Answer: 3/8, 0.405, 7/16, 0.45

Question 12 *Practice question written for this worksheet (not from a past paper)*

Write $0.\dot{7}$ as a fraction.

$x = 0.777\ldots,\ 10x = 7.777\ldots,\ 9x = 7$

$x = \frac{7}{9}$

Answer: 7/9

Question 13 *Practice question written for this worksheet (not from a past paper)*

Write $0.\dot{1}\dot{8}$ as a fraction in its simplest form.

$x = 0.1818\ldots,\ 100x = 18.1818\ldots,\ 99x = 18$

$x = \frac{18}{99} = \frac{2}{11}$

Answer: 2/11

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Question 14 *Practice question written for this worksheet (not from a past paper)*

Pens cost £1.35 each. Sam buys 12 pens.

He pays with a £20 note.

Work out how much change he should get.

Cost: 12 × £1.35 = £16.20

Change: £20 − £16.20 = £3.80

Answer: 3.80

Question 15 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 15*

15		eg $(10\,000x =)$ 7636.36… $(100x =)$ 76.36… **or** $(1000x =)$ 763.63… $(10x =)$ 7.63… **or** $(100x =)$ 76.363… $(x =)$ 0.763… oe		2	M1	For 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 75.6 or 756 or 7560 etc) eg $(10\,000x =)$ 7636.36... and $(100x =)$ 76.36.... **or** $(1000x =)$ 763.63… and $(10x =)$ 7.63… **or** $(100x =)$ 76.363… and $(x =)$ 0.763… with intention to subtract. (if recurring not shown then showing at least **one** of the numbers to at least 5sf) or $\frac{7}{10}+1000x(63.6\dot{3})-10x(0.6\dot{3})$
		eg $10\,000x - 100x = 7636.36... - 76.36... = 7560$ **and** $\frac{7560}{9900}=\frac{42}{55}$ **or** $1000x - 10x = 763.63... - 7.63... = 756$ $(990x = 756)$ **and** $\frac{756}{990}=\frac{42}{55}$ **or** $100x - x = 76.363... - 0.763... = 75.6$ **and** $\frac{75.6}{99}=\frac{42}{55}$ $(99x = 75.6)$ **or** $\frac{7}{10}+\frac{63}{990}=\frac{7\times99+63}{990}=\frac{42}{55}$	shown		A1	for completion to $\frac{42}{55}$ dep on M1 and must use algebra for this final mark to be awarded [allow for instance $99x = 75.6$ and then $\frac{756}{990}=\frac{42}{55}$] **No algebra used gets a maximum of 1**
		Working required				**Total 2 marks**

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2019 · Question 15

15		$x = 0.25454...$ $100x = 25.454...$ $10x = 2.5454...$ $1000x = 254.54...$		2	M1	For 2 recurring decimals that when subtracted give a whole number or terminating decimal eg 25.2 or 252 etc eg $100x = 25.454...$ and $x = 0.25454...$ **or** $1000x = 254.54...$ and $10x = 2.5454...$ with intention to subtract. (if recurring dots not shown then showing at least the digits 25454, ie 5sf) **or** $0.2 + 0.05\dot{4}$ **and** eg $x = 0.05454..., 100x = 5.4545...$ with intention to subtract.
		eg $100x - x = 25.454... - 0.254... = 25.2$ **and** $\frac{25.2}{99} = \frac{14}{55}$ **or** $1000x - 10x = 254.545... - 2.545... = 252$ **and** $\frac{252}{990} = \frac{14}{55}$ **or** $100x - x = 5.4545... - 0.05454... = 5.4$ **and** $\frac{5.4}{99} = \frac{54}{990}\left(= \frac{3}{55}\right)$ **and** $\frac{2\times99+54}{990} = \frac{252}{990} = \frac{14}{55}$ or $\frac{5.4}{99} = \frac{54}{990} = \frac{3}{55}$ and $\frac{11+3}{55} = \frac{14}{55}$	show		A1	for completion to $\frac{14}{55}$
						Total 2 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 16

16	eg $(100x =)\ 37.878\ldots$ $(x =)\ 0.378\ldots$ **or** $(1000x =)\ 378.78\ldots$ $(10x =)\ 3.78\ldots$ **or** $(100000x =)\ 37878.78\ldots$ $(1000x =)\ 378.78\ldots$		2	M1 for 2 **correct** recurring decimals that when subtracted should give a whole number or terminating decimal with intention to subtract. (ie may give 37.5 or 375 or 37500 or 78 etc) eg $(1000x =)\ 37.878\ldots$ and $(x =)\ 0.378\ldots$ **or** $(1000x =)\ 378.78\ldots$ and $(10x =)\ 3.78\ldots$ **or** $(100000x =)\ 37878.78\ldots$ and $(1000x =)\ 378.78\ldots$ **or** $0.3 + 0.07\dot{8}$ **and** $(1000y =)\ 78.787...$ and $(10y =)\ 0.78....$ with intention to subtract. x is not required to award this mark (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least **5sf**) **NB** Accept bar notation for dot notation to indicate recurring decimals
	eg $100x - x = 37.878\ldots - 0.378\ldots = 37.5$ **and** $\frac{37.5}{99} = \frac{25}{66}$ **or** $1000x - 10x = 378.78... - 3.78... = 375$ **and** $\frac{375}{990} = \frac{25}{66}$ **or** $100000x - 1000x = 37878.78... - 378.78... = 37500$ **and** $\frac{37500}{99000} = \frac{25}{66}$ **or** $1000y - 10y = 78.787... - 0.7878... = 78$ **and** $0.3 + \frac{78}{990} = \frac{0.3\times990+78}{990} = \frac{375}{990} = \frac{25}{66}$ oe	Shown		A1 for completion to $\frac{25}{66}$ dep on M1and must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required			**Total 2 marks**

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Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 16*

Q	Working	Answer	Mark	Notes
	(1000x =) 612.12..._ eg (10x=) 6.12... **or** (1000x =) 612.12..._ (10x=) 6.12... **or** (100x =) 61.212..._ (x=) 0.612...		2	M1 for 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 60.6 or 606 or 6060 etc) eg (1000x =) 612.12... and (10x =) 6.12.... **or** (100 000x =) 61 212.12.... and (1000x =) 612.... **or** (100x =) 61.212.... and (x =) 0.612.... with intention to subtract x is not required to award this mark (if recurring dots not shown in **both** numbers then showing at least **one** of the numbers to at least 5sf) or $\frac{6}{10}+1000x(12.12)-10x(0.12)$
	eg $10000x - 100x =$ 6121.21… − 61.21…=6060 (9900x = 6060) **and** $\frac{6060}{9900}=\frac{101}{165}$ **or** $1000x - 10x = 612.12... - 6.12... = 606$ (990x = 606) **and** $\frac{606}{990}=\frac{101}{165}$ **or** $100x - x = 61.212... - 0.612...$ 60.6 (99x = 60.6) **and** $\frac{60.6}{99}=\frac{101}{165}$ oe **OR** 0.6 +… and (1000x −100x = 990x = 12) **and** $0.6+\frac{12}{990}=\frac{0.6\times990+12}{990}=\frac{101}{165}$ oe	shown		A1 for completion to $\frac{101}{165}$ dep on M1 and must use algebra for this final mark to be awarded [allow for instance 99x = 60.6 and then $\frac{606}{990}=\frac{101}{165}$] **No algebra used gets a maximum of 1 mark**
	Working required			**Total 2 marks**

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Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 16

16		eg $10\,000x = 1767.67\ldots$ $100x = 17.67\ldots$ **or** $1000x = 176.76\ldots$ $10x = 1.76\ldots$ **or** $100x = 17.676\ldots$ $x = 0.176\ldots$ oe		2	M1	for 2 recurring decimals that when subtracted give a whole number or terminating decimal (17.5 or 175 or 1750 etc) eg $10\,000x = 1767.67...$ and $100x = 17.676....$ **or** $1000x = 176.76\ldots$ and $10x = 1.7676\ldots$ **or** $100x = 17.676\ldots$ and $x = 0.17676\ldots$ with intention to subtract. (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 5sf) **or** $0.1 + 0.0\dot{7}\dot{6}$ **and** eg $100x = 7.6767...$, $x = 0.07676...$ with intention to subtract.
		eg $10\,000x - 100x = 1767.67... - 17.67... = 1750$ **and** $\frac{1750}{9900} = \frac{35}{198}$ **or** $1000x - 10x = 176.76... - 1.76... = 175$ **and** $\frac{175}{990} = \frac{35}{198}$ **or** $100x - x = 17.676... - 0.176... = 17.5$ **and** $\frac{17.5}{99} = \frac{35}{198}$ **or** eg $10x - x = 7.6767... - 0.07676... = 7.6$ **and** $0.1 + \frac{7.6}{99} = \frac{0.1\times99+7.6}{99} = \frac{17.5}{99} = \frac{35}{198}$ oe	shown		A1	for completion to $\frac{35}{198}$ dep on M1
		Working required				**Total 2 marks**

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 17

17	eg $(100x =)\ 53.939\ldots$ $(x =)\ 0.539\ldots$ **or** $(1000x =)\ 539.39\ldots$ $(10x =)\ 5.39\ldots$ **or** $(100\,000x =)\ 539\,39.39\ldots$ $(1000x =)\ 539.39\ldots$ **or** $0.5 + 0.0\dot{3}\dot{9}$ **and** eg $(1000y =)\ 39.393\ldots$ $(10y =)\ 0.39\ldots$		2	M1	for 2 **correct** recurring decimals that when subtracted should give a whole number or terminating decimal **and** intention to subtract (ie may give 53.4 or 534 or 53 400 or 39 or 3900 etc) eg any of the following with intention to subtract $(100x =)\ 53.939...$ and $(x =)\ 0.539....$ **or** $(1000x =)\ 539.39...$ and $(10x =)\ 5.39....$ **or** $(100\,000x =)\ 53939....$ and $(1000x =)\ 539.39....$ **or** $0.5 + 0.0\dot{3}\dot{9}$ **and** $(1000y =)\ 39.393\ldots$ and $(10y =)\ 0.39\ldots$ x is not required to award this mark if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 5sf is sufficient NB Accept bar notation for dot notation to indicate recurring decimals
	$100x - x = 53.939\ldots - 0.539\ldots = 53.4$ **and** $\frac{53.4}{99} = \frac{89}{165}$ **or** $1000x - 10x = 539.39\ldots - 5.39\ldots = 534$ **and** $\frac{534}{990} = \frac{89}{165}$ **or** $100\,000x - 1000x = 53939 - 539 = 53\,400$ **and** $\frac{53\,400}{99\,000} = \frac{89}{165}$ **or** $1000y - 10y = 39.393\ldots - 0.39\ldots = 39$ **and** $0.5 + \frac{39}{990} = \frac{0.5\times990+39}{990} = \frac{534}{990} = \frac{89}{165}$	Shown		A1	dep on M1, for completion to $\frac{89}{165}$ must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required				**Total 2 marks**

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 17

17	eg $x = 0.34545....$ **and** $100x = 34.545....$ with intention to subtract **OR** $10x = 3.4545....$ **and** $1000x = 345.45....$ with intention to subtract Must include algebra as the question asked for 'using algebra'		2	M1	for 2 recurring decimals (they must identify or show the pair they are using) that when subtracted give a whole number or terminating decimal eg $100x = 34.545....$ and $x = 0.34545....$ **OR** $1000x = 345.45....$ and $10x = 3.4545....$ with intention to subtract. (If recurring dots not shown then showing at least the digits 34545, i.e. 5sf for **one** of the numbers that they are using) **OR** $0.3 + 0.0454545....$ and Eg $10x = 0.454545...$ and $1000x = 45.4545..$
	eg $100x - x = 34.545.... - 0.34545....$ **and** $99x = 34.2$ **and** $\frac{34.2}{99} = \frac{19}{55}$ oe **OR** $1000x - 10x = 345.45.... - 3.4545....$ **and** $990x = 342$ **and** $\frac{342}{990} = \frac{19}{55}$ oe **OR** $0.3 + ...$ and $(1000x - 100x = 990x = 45)$ and $0.3 + \frac{45}{990} = \frac{3\times99+45}{990} = \frac{19}{55}$ oe	shown		A1	for completion to $\frac{19}{55}$
					Total 2 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2012 · Question 17

17.	$x = 0.1777....$ and $10x = 1.777..$ $9x = 1.6$	16/90 oe		See at least 3 sevens or recurring symbol. Condone omission of x. M1 Accept $10x = 1.777..$ and $100x = 17.77..$ A1 Must be integers in numerator and denominator but not 8 & 45 N.B for $0.1777 = 1/10 + 0.0777..$ (0.777 needs to be shown to be 7/90 to gain first M1)
				Total 2 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 18

18	$(1000x =)$ 702.702..._ eg $(x =)$ 0.702... $(10000x =)$ 7027.02..._ **or** $(10x =)$ 7.02... $(100000x =)$ 70270.20..._ **or** $(100x =)$ 70.20...		2	M1	for 2 recurring decimals that when subtracted give a whole number or terminating decimal with intention to subtract. (ie give 702 or 7020 or 70200 etc) eg $(1000x =)$ 702.702... and $(x =)$ 0.702.... **or** $(10000x) = 7027.02....$ and $(10x) = 7.02....$ **or** $(100000x =)$ 70270.20.... and $(100x =)$ 70.20.... with intention to subtract x is not required to award this mark (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 6sf) NB Accept bar notation for dot notation to indicate recurring decimals
	eg $1000x - x = 702.70... - 0.702... = 702$ $(999x = 702)$ **and** $\frac{702}{999} = \frac{26}{37}$ **or** $10000x - 10x = 7027.02... - 7.02... = 7020$ $(9990x = 7020)$ **and** $\frac{7020}{9990} = \frac{26}{37}$ **or** $100000x - 100x = 70270.20... - 70.20 = 70200$ $(99900x = 70200)$ **and** $\frac{70200}{99900} = \frac{26}{37}$	shown		A1	for completion to $\frac{26}{37}$ dep on M1 and must use algebra for this final mark to be awarded **No algebra used gets a maximum of 1 mark**
	Working required				**Total 2 marks**

Question 24 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 18*

18	eg $1000x = 306.306\ldots$ $x = 0.306\ldots$ OR eg $1000\,000x = 306\,306.(\ldots)$ $1000x = 306.306\ldots$		2	M1	M1 for two correct **algebraic equations** involving recurring decimals that when subtracted give a whole number or terminating decimal (306 or 306 000 etc) with intention to subtract. eg $1000x = 306.306\ldots$ and $x = 0.306\ldots$ **or** $1000\,000x = 306306.(\ldots)$ and $1000x = 306.306$ (if recurring dots not shown in both numbers then showing at least **one** of the numbers to at least 6sf)
	eg $1000x - x = 306.306\ldots - 0.306306\ldots = 306$ **and** $\frac{306}{999} = \frac{34}{111}$ or $999x = 306$ **and** $\frac{306}{999} = \frac{34}{111}$ OR eg $1000\,000x - 1000x = 306306.(\ldots) - 306.306 = 306\,000$ **and** $\frac{306\,000}{999\,000} = \frac{34}{111}$ or $999\,000x = 306\,000$ **and** $\frac{306\,000}{999\,000} = \frac{34}{111}$	shown		A1	for completion to $\frac{34}{111}$ dep on M1
	Working required				**Total 2 marks**

Question 25 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2011 · Question 18*

18.	$x = 0.396396\ldots$ $1000x = 396.396\ldots$ $999x = 396$	$\frac{44}{111}$	2	M1 A1 must reach $\frac{396}{999}$ or equivalent fraction (but not $\frac{44}{111}$)
				Total 2 marks

Question 26 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 20*

20.	$(x =)\ 0.01515..$ and $(100x =)\ 1.515\ldots$ $99x = 1.5$ 15/990 oe	1/ 66 *	2	M1 or $10x = 0.1515..$and $1000x = 15.1515..$ selected A1 but not 1/66 Numerator and denominator have to be integers. * Answer given
				Total 2 marks

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CHALLENGE QUESTIONS

Question 27 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 3*

Q	Working	Answer	Mark	Notes
(a)		14 squares shaded	1	B1 any 14 squares shaded
(b)		two tenths	1	B1 allow 'tenths' or $\frac{2}{10}$
(c)		0.625	1	B1
(d)		2.008, 2.081, 2.8, 2.803, 2.83	2	B2 for all numbers in correct order (B1 for one number when covered leaves the others in order or for all numbers in correct reverse order)
				Total 5 marks

Question 28 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 13*

13	e.g. $x = 0.68\dot{1}\dot{1}$ and $100x = 68.\dot{1}\dot{8}$ **or** $10x = 6.8\dot{1}\dot{1}$ and $1000x = 681.\dot{8}\dot{1}\dot{1}$			M1 e.g. two decimals that when subtracted give a finite decimal (must show understanding of recurring figures by 'dot' or at least 2 lots of 18 or 81 after the decimal point). Algebra required, use of any letter.
	$99x = 67.5$, $x = \frac{67.5}{99} = \frac{15}{22}$ **or** $990x = 675$, $x = \frac{675}{990} = \frac{15}{22}$ oe	show	2	A1 dep for completing the 'show that' arriving at given answer from correct working.
				Total 2 marks

Question 29 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 16*

16		eg $10\,000x = 2813.13\ldots$ $100x = 28.13\ldots$ **or** $1000x = 281.313\ldots$ $10x = 2.813\ldots$ **or** $100x = 28.1313\ldots$ $x = 0.2813\ldots$ oe		2	M1 For 2 recurring decimals that when subtracted give a whole number or terminating decimal (27.85 or 278.5 or 2785 etc) eg $10\,000x = 2813.13\ldots$ and $100x = 28.1313\ldots$ **or** $1000x = 281.313\ldots$ and $10x = 2.81313\ldots$ **or** $100x = 28.1313\ldots$ and $x = 0.281313\ldots$ with intention to subtract. (if recurring dots not shown then showing at least **one** of the numbers to at least 6sf) **or** $0.28 + 0.00\dot{1}\dot{3}$ **and** eg $100x = 0.1313\ldots$, $10000x = 13.1313\ldots$ with intention to subtract.
		eg $10\,000x - 100x = 2813.13\ldots - 28.1313\ldots = 2785$ **and** $\frac{2785}{9900} = \frac{557}{1980}$ **or** $1000x - 10x = 281.313\ldots - 2.81313\ldots = 278.5$ **and** $\frac{278.5}{990} = \frac{557}{1980}$ **or** $100x - x = 28.1313\ldots - 0.281313\ldots = 27.85$ **and** $\frac{27.85}{99} = \frac{557}{1980}$ **or** eg $10\,000x - 100x = 13.1313\ldots - 0.1313\ldots = 13$ **and** $0.28 + \frac{13}{9900} = \frac{28\times99+13}{9900} = \frac{2785}{9900} = \frac{557}{1980}$ oe	shown		A1 for completion to $\frac{557}{1980}$ dep on M1 *(NB: this is a "use algebra to show that…" question, so we need to see algebra as well as seeing all the stages of working to award full marks)*
					Total 2 marks

Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 16

Question	Working	Answer	Mark	Notes
16	eg $1000x = 438.38\ldots$ $10x = 4.38\ldots$ **or** $100x = 43.838\ldots$ $x = 0.438\ldots$ oe		2	M1 For **selecting** 2 correct recurring decimals that when subtracted give a whole number or terminating decimal (43.4 or 434 etc) eg $1000x = 438.38...$ and $10x = 4.38....$ **or** $100x = 43.838\ldots$ and $x = 0.438\ldots$ with intention to subtract. (if recurring dots not shown then showing at least **one** of the numbers to at least 5sf) **or** $0.4 + 0.0\dot{3}\dot{8}$ **and** eg $1000x = 38.38...$ & $10x = 0.3838...$, with intention to subtract.
	eg $1000x - 10x = 438.38\ldots - 4.38\ldots = 434$ **and** $\frac{434}{990} = \frac{217}{495}$ **or** eg $100x - x = 43.838\ldots - 0.438\ldots = 43.4$ **and** $\frac{43.4}{99} = \frac{217}{495}$ **or** eg $1000x - 10x = 38.38\ldots - 0.3838 = 38$ and $0.4 + \frac{38}{990} = \frac{4\times99+38}{990} = \frac{434}{990} = \frac{217}{495}$ oe *working required*	Clearly shown		A1 For completion to $\frac{217}{495}$ dep on M1 and use of some algebra
				Total 2 marks

Question 31

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 18

18	E.g. $y = 0.4x....$ **and** $10y = 4.x....$ $(10y - y = 4.x - 0.4$ oe) **or** $10y = 4.x....$ **and** $100y = 4x.x....$ $(100y - 10y = 4x - 4$ oe) **or** $100y = 4x.x....$ **and** $1000y = 4xx.x....$ $(1000y - 100y = 4xx - 4x$ oe)		3	M1 for selecting 2 correct recurring decimal expressions and then a demonstration to subtract (If recurring dots not shown then allow each expression to 1 dp e.g. $y = 0.4x....$ **and** $100y = 4x.x....$ as a pair **and** $100y - y$ or $4x.x.... - 0.4x....$) or an answer of $y = \frac{4x-4}{90}$ oe
	E.g. $9y = 4\frac{x}{10} - \frac{4}{10} = \frac{40+x-4}{10}$ oe or $90y = 40 + x - 4$ oe or $900y = 400 + 10x + x - 40 - x$ oe			M1 for a correct subtraction with correct expressions simplified
	Working required	$\frac{36+x}{90}$		A1 dep on M2 oe
				Total 3 marks